Artificial Intelligence

Key Concepts and Problem Solving

A Comprehensive Study Guide

1. Water Jug Problem

1.1 Introduction

The Water Jug Problem is a classic AI problem used to demonstrate state space search. It involves two jugs of different capacities with no measurement markings, and the goal is to measure a specific amount of water using only these jugs.

1.2 Problem Statement

Given: A 4-litre jug and a 3-litre jug (both unmarked).

Goal: Measure exactly 2 litres of water in the 4-litre jug.

Allowed Operations: Fill a jug, Empty a jug, Pour water from one jug to another.

1.3 Problem Formulation

State: (x, y) where x = water in 4L jug, y = water in 3L jug

Initial State: (0, 0) — both jugs are empty

Goal State: (2, 0) — 2 litres in the 4L jug

Operators: 6 possible operations on the jugs

1.4 Solution Steps

Step	4L Jug (x)	3L Jug (y)	Action
1	0	0	Initial State
2	0	3	Fill 3L jug
3	3	0	Pour 3L jug into 4L jug
4	3	3	Fill 3L jug again
5	4	2	Pour from 3L to 4L (4L becomes full)
6	0	2	Empty the 4L jug
7	2	0	Pour 2L from 3L jug to 4L jug — GOAL!

1.5 Key Points

- It demonstrates state space search in AI.
- Each state is represented as (x, y).
- BFS or DFS can be used to find the solution.
- Multiple solutions may exist with different path lengths.

2. Vacuum Cleaner Problem

2.1 Introduction

The Vacuum Cleaner Problem is one of the simplest and most commonly used examples to explain the concept of an AI agent. It is used to demonstrate how an agent perceives its environment and acts upon it to achieve a goal.

2.2 Problem Description

Environment: Two rooms: Room A and Room B.

Agent: A vacuum cleaner robot.

Percepts: Current location (A or B) and whether the current room is dirty or clean.

Actions: Move Left, Move Right, Suck (clean the dirt).

Goal: Clean all the dirt in both rooms.

2.3 State Space

State	Room A	Room B	Vacuum Location
1	Dirty	Dirty	A
2	Dirty	Dirty	B
3	Clean	Dirty	A
4	Clean	Dirty	B
5	Dirty	Clean	A
6	Dirty	Clean	B
7	Clean	Clean	A
8 (Goal)	Clean	Clean	B

2.4 Agent Behaviour Rules

- If current room is dirty → Suck (clean the room).
- If current room is clean and in Room A → Move Right to Room B.
- If current room is clean and in Room B → Move Left to Room A.
- If both rooms are clean → Stop (Goal Achieved).

2.5 PEAS Description

Component	Description
Performance Measure	Number of rooms cleaned, time taken, energy consumed
Environment	Two rooms (A and B) which can be dirty or clean
Actuators	Suction mechanism, wheels to move left or right
Sensors	Location sensor, dirt sensor

2.6 Key Points

- It is an example of a simple reflex agent.
- The environment is fully observable, deterministic and discrete.
- It is used to introduce concepts of agent, environment and actions.
- Performance measure can include number of clean rooms and time taken.

3. 8-Puzzle Problem

3.1 Introduction

The 8-Puzzle Problem is a classic AI sliding puzzle problem. It consists of a 3x3 grid with 8 numbered tiles (1 to 8) and one blank space. The objective is to rearrange the tiles from an initial configuration to reach the goal configuration by sliding tiles into the blank space.

3.2 Problem Formulation

State: A 3x3 grid arrangement of 8 numbered tiles and one blank tile.

Initial State: Any random arrangement of tiles.

Goal State: Tiles arranged in order: 1,2,3 / 4,5,6 / 7,8,Blank.

Operators: Move blank tile Up, Down, Left, or Right.

Path Cost: Number of moves made to reach the goal.

3.3 Example

Initial State: **Goal State:**

2	8	3
1	6	4
7		5

1	2	3
4	5	6
7	8	

3.4 Search Algorithms Used

- Breadth First Search (BFS) — guarantees optimal solution but uses more memory.
- Depth First Search (DFS) — uses less memory but may not find optimal solution.
- A* Algorithm — uses heuristic function h(n) to find optimal solution efficiently.
- Hill Climbing — fast but may get stuck at local maxima.

3.5 Heuristic Functions for 8-Puzzle

Heuristic	Description
h1 - Misplaced Tiles	Count of tiles not in their goal position
h2 - Manhattan Distance	Sum of distances of each tile from its goal position

Heuristic	Description
h2 is more accurate	Manhattan distance gives better estimate than misplaced tiles

3.6 Key Points

- Total possible states = 9! = 362880, but only half are reachable.
- A* with Manhattan distance heuristic is most efficient.
- It is used to demonstrate search strategies in AI.
- Branch factor = 2 to 4 (blank tile has 2 to 4 possible moves).

4. Problem Solving Agent

4.1 Definition

A Problem Solving Agent is a type of goal-based agent that decides what to do by finding sequences of actions that lead to a desired goal state. It uses search algorithms to explore possible action sequences and selects the best one.

4.2 Working of a Problem Solving Agent

- Step 1 — Goal Formulation: Define what the agent wants to achieve.
- Step 2 — Problem Formulation: Define states, actions, and costs.
- Step 3 — Search: Find a sequence of actions that leads to the goal.
- Step 4 — Execute: Perform the actions in the found solution.

4.3 Components of Problem Formulation

Component	Description	Example (8-Puzzle)
Initial State	Starting configuration of the problem	Random tile arrangement
Goal State	Desired final configuration	Tiles in order 1-8
Actions / Operators	Possible moves the agent can make	Move blank Up/Down/Left/Right
Path Cost	Cost of reaching a state from initial state	Number of moves made
State Space	Set of all possible states	All 362880 configurations

4.4 Properties of a Good Problem Solving Agent

- Completeness — Always finds a solution if one exists.
- Optimality — Finds the least cost solution.
- Time Efficiency — Finds solution in reasonable time.
- Space Efficiency — Uses reasonable amount of memory.

4.5 Difference: Problem Solving Agent vs Simple Reflex Agent

Feature	Problem Solving Agent	Simple Reflex Agent
Decision Making	Plans a sequence of actions	Reacts to current percept only
Memory	Uses internal state and search	No memory used
Goal	Works towards a defined goal	No specific goal

Feature	Problem Solving Agent	Simple Reflex Agent
Example	Chess playing program	Thermostat

5. Types of AI Agents

5.1 Introduction

An AI agent is an entity that perceives its environment through sensors and acts upon it through actuators. Based on their decision-making capability and use of internal state, AI agents are classified into five types.

5.1 Simple Reflex Agent

A simple reflex agent selects actions based only on the current percept, ignoring all past history. It works on condition-action rules (if-then rules).

Working: Perceives current state → Matches condition → Executes action.

Example: Thermostat, Traffic light controller.

Limitation: Cannot handle partially observable environments.

5.2 Model-Based Reflex Agent

A model-based reflex agent maintains an internal model (state) of the world. It uses both the current percept and the internal state to decide its action. It can handle partially observable environments.

Working: Perceives → Updates internal state → Matches condition → Executes action.

Example: Robot vacuum cleaner with memory of visited rooms.

Advantage: Can work in partially observable environments.

5.3 Goal-Based Agent

A goal-based agent works towards achieving a specific goal. It considers future actions and chooses those that lead to the goal. It uses search and planning algorithms.

Working: Perceives → Considers goal → Plans actions → Executes.

Example: Navigation system finding best route to destination.

Advantage: More flexible, can change behavior when goal changes.

5.4 Utility-Based Agent

A utility-based agent selects actions that maximize a utility function. When multiple goal states exist, it picks the one with the highest utility (happiness). It handles conflicting goals and trade-offs.

Working: Perceives → Evaluates utility of each action → Executes best action.

Example: Self-driving car balancing speed, safety, and fuel efficiency.

Advantage: Makes best decision among multiple choices.

5.5 Learning Agent

A learning agent improves its performance over time based on experience. It has a learning element that modifies the agent's behavior based on feedback from the performance element.

Components: Learning Element, Performance Element, Critic, Problem Generator.

Example: Spam filter that learns from user feedback.

Advantage: Can operate in unknown environments and improve over time.

5.6 Comparison of All Agent Types

Agent Type	Uses Memory	Uses Goal	Uses Utility	Learns
Simple Reflex	No	No	No	No
Model-Based Reflex	Yes	No	No	No
Goal-Based	Yes	Yes	No	No
Utility-Based	Yes	Yes	Yes	No
Learning Agent	Yes	Yes	Yes	Yes

5.7 Key Points

- Simple reflex agents are the simplest and least powerful.
- Model-based agents can handle partial observability.
- Goal-based agents plan sequences of actions.
- Utility-based agents make optimal decisions among multiple choices.
- Learning agents are the most powerful and adaptable.

6. Summary

Topic	Key Concept	AI Technique Used
Water Jug Problem	State space search with operations on jugs	BFS / DFS
Vacuum Cleaner Problem	Simple reflex agent in 2-room environment	Simple Reflex Agent
8-Puzzle Problem	Sliding tile puzzle with heuristic search	A* / BFS / DFS
Problem Solving Agent	Goal-based agent using search and planning	Search Algorithms
Types of Agents	5 types based on intelligence level	Various AI Techniques